SUSTAINABILITY

Built to last, built to be fixed.

ExoRehab is designed so rehabilitation technology is affordable, repairable and kinder to the planet.

Recycled materials
3D-printed recycled PETG frame and washable, replaceable cuffs.

Modular & repairable
Cuffs, battery and actuator snap apart — fix a part, not the device.

Low-power electronics
Edge AI on an ESP32, deep sleep between sessions, USB-C charging.

Fewer hospital trips
Remote monitoring brings therapy home.

Supporting the SDGs

- 3 Good Health & Well-being
- 9 Industry & Innovation
- 10 Reduced Inequalities
- 12 Responsible Consumption

THE TEAM

Students from Surakarta.

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WORKING PAPER TITLE

ExoRehab: A Sustainable Smart Exoskeleton for Post-Stroke Hemiparesis Rehabilitation with AI-Driven Adaptive Assistance and Real-Time IoT Monitoring

Key features

- AI Adaptive Assistance
- Real-Time IoT Monitoring
- Sustainable by Design
- Home-Based Therapy
- Progress Analytics
- Safety First

ExoRehab is a student research prototype and concept, not a certified medical device. App data shown is simulated. Title and patient criteria are still being refined.

ExoRehab

EXOSKELETON REHABILITATION



Smart exoskeleton for post-stroke recovery.

AI-driven adaptive assistance · Real-time IoT monitoring · Sustainable design

SMA Negeri 1 Surakarta
SMA Negeri 3 Surakarta

01 · THE PROBLEM

Stroke takes away movement. Therapy is hard to reach.

Hemiparesis — weakness on one side of the body — is the most common motor effect of stroke. Recovery needs hundreds of guided repetitions, but therapists, clinic time and travel are limited, especially once patients go home.

12.2M — new strokes worldwide every year

~80% — of survivors live with motor weakness on one side

#1 — cause of death in Indonesia is stroke

Sources: World Stroke Organization, 2022 · Hemiparesis prevalence, stroke literature · Indonesian Ministry of Health / IHME

Our answer

A lightweight upper-limb exoskeleton that **follows the patient's intention**, gives only the help that is missing, and keeps the care team informed — at home.

02 · HOW IT WORKS

Sense, think, assist, monitor.

1 Sense
sEMG, IMU and force sensors read muscle intent, joint angle and grip in real time.

2 Think
An on-device AI model estimates patient effort and fatigue every 50 ms.

3 Assist
The actuator adds only the torque the patient is missing — assist-as-needed.

4 Monitor
ESP32 streams sessions over IoT to therapists and family dashboards.

03 · AI ADAPTIVE ASSISTANCE

$$\tau_{motor} = (1 - \hat{e}) \cdot \tau_{task}$$

ê = patient effort estimated from sEMG + joint motion

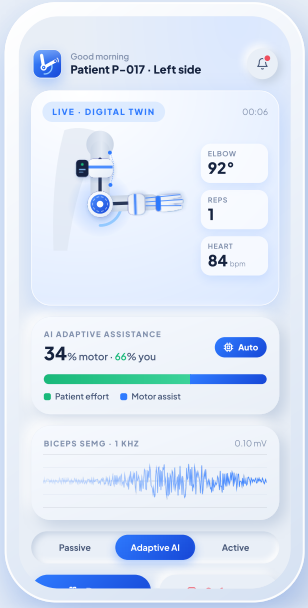
As the patient gets stronger, motor help **fades automatically**. Fatigue or spasm raises assistance or triggers a soft stop.

Safety first
Mechanical joint limits, spasm detection and an instant soft-stop button protect the patient.

Upper-limb focus
Elbow flexion–extension with a tendon-assisted glove for grip practice.

04 · THE APP

Every session, live & measurable.



- Live digital twin
- sEMG muscle signal
- AI assist vs. effort
- Heart rate & reps
- Weekly progress
- Therapist alerts

Real-time IoT
ESP32 → MQTT/TLS cloud → therapist & family dashboards.

Progress analytics
Range of motion, effort and consistency trends in weekly reports.

Try the live demo

Open the ExoRehab web app — demo data runs right in your browser.